Julian Hidalgo, Jose Avila, Mauricio Martinez, Julio Franco,
CSCI 164

Summary Report

Project: Applied Machine Learning with scikit-learn

Project Overview:

Our project was to compare different supervised learning models across three datasets with the use of scikit-learning. We are supposed to apply a full machine learning workflow on different problems instead of solely relying on one dataset. The datasets we used consisted of classification and regression. The three datasets we decided to use from Kaggle are Spotify Yearly(2014-2026), Heart Disease, and Housing Prices. This combination of datasets gives us a spread of difficulty, types, and methods to use.

Datasets and Preprocessing

1. Heart Disease Dataset

This dataset comes from 1988, with four databases combined: Cleveland, Hungary, Switzerland, and Long Beach V. It has 1,025 patients (with deduplication), and has no missing values. It has a close balance binary classification, and has many features like age, sex, cholesterol, and more. For prepossing, there was duplication, so we have to remove those. Since it had no missing values, we changed nothing, used Logistic Regression, k-NN, and Naive Bayes.

2. Spotify Yearly Track Metadata (2014–2026)

This dataset has the metadata that was collected using the Spotify Web API. It has 12,397 tracks, and it also has the features of year, duration, release, artists, and name. With 5 classifications: love, night, remix, rock, and sad. For prepossing, we removed tracks that go over 15 minutes, and combined the files into one CSV. Had to remove one file as it was a duplicate of one of the previous files in the dataset.

3. Housing Prices Dataset

This dataset is trying to predict housing prices based on certain things like house area, bedrooms, access to a main road, and other things. For preprocessing, there was no duplication throughout the dataset, no missing values, and we had to change around six columns with yes/no to integers as 1/0, as well as the furnishing status column to 1/0.

Models Used

For the Heart Disease dataset, we used k-NN, Logistic Regression, and Naive Bayes. For the Housing Prices Dataset, we used linear Regression, k-NN, and Ridge Regression. For the Spotify Yearly Track Metadata (2014–2026). We used Logistic Regression, Decision Tree, and k-NN. These models gave us the ability to compare linear models, tree-based models, and local neighborhood models, with their respective databases.

Validation and Hyperparameter Tuning

All of the models were evaluated using an 80/20 stratified or a random train/test split with an additional 5-fold cross-validation on the training set. We used GridSearchCV with a 5-fold stratified cross-validation on all tunable models (k-NN and Decision Tree). Linear Regression and Naive Bayes don't have any or no meaningful hyperparameters, so we used them as baselines.

Results:

Heart Disease — Classification Results

Model	Test Acc	F1-Score	ROC-AUC	CV Mean	CV Std
k-NN (tuned)	0.8197	0.8358	0.8945	0.8504	0.0468
Logistic Regression (tuned)	0.8033	0.8235	0.8712	0.8379	0.0861
Naive Bayes	0.7869	0.7937	0.8842	0.8172	0.0489
Decision Tree (untuned)	0.8033	0.8182	0.8019	0.7801	0.0280
Decision Tree (tuned)	0.7049	0.7273	0.7971	0.8094	0.0570

Best model k-NN (tuned): No Disease — Precision 0.81, Recall 0.79, F1 0.80 | Disease — Precision 0.83, Recall 0.85, F1 0.84

Spotify — Classification Results

Model	Test Acc	Weighted F1	CV Mean Acc	CV Std
k-NN (tuned)	0.3726	0.3652	0.3663	0.0048
Decision Tree (tuned)	0.3694	0.3522	0.3665	0.0107
k-NN (k=11, untuned)	0.3605	0.3523	0.3596	0.0110
Logistic Regression (tuned)	0.3524	0.3164	0.3594	0.0138

Per-class F1 (best k-NN): love=0.32, night=0.28, remix=0.61, rock=0.34, sad=0.27. The remix category was the most distinguishable from the metadata features available.

Spotify — Regression Results

Model	MAE (min)	RMSE (min)	Test R^2	CV R^2
Decision Tree (tuned) ★	0.768	1.104	0.0774	0.0821
Linear Regression	0.776	1.112	0.0632	0.0712
Decision Tree (untuned)	0.802	1.156	-0.0130	-0.0704

Housing Prices — Regression Results

Model	MAE (₹)	RMSE (₹)	Test R^2	CV R^2	CV Std
Linear Regression	979,680	1,331,071	0.6495	0.6486	0.0469
Ridge (untuned)	979,549	1,331,290	0.6494	0.6487	0.0466
Ridge (tuned α=50)	978,706	1,346,128	0.6415	0.6503	0.0338
k-NN Regressor (tuned)	1,034,828	1,450,198	0.5839	0.6366	0.0333
Decision Tree (tuned)	1,258,021	1,689,677	0.4352	0.4533	0.1995

Comparison of Prior Work

Dataset	Prior Work Metric	Our Best Result	Gap / Notes
Heart Disease	PMC 2025: 99% Accuracy (RF deep learning)	k-NN: 81.97% Acc, AUC 0.895	−17% Acc; deep learning vs. shallow classifiers
Spotify	Mansisu08 2023: 47.27% LR Acc (audio features)	k-NN: 37.26% Acc (metadata only)	−10% Acc; no audio features available
Housing	Madhuri et al. 2019: LR R^2≈0.678	LR: R^2=0.6495	−0.028; consistent conclusions, different dataset

Interpretation of Results

- Heart Disease: The k-NN model (81.97% accuracy, ROC-AUC 0.895) showed strong discrimination. The 17-point gap vs. prior work reflects the expected performance ceiling of shallow classifiers relative to deep learning ensembles. Notably, both studies agree on the most predictive features: chest pain type (cp), ST depression (oldpeak), and exercise-induced angina.
- Spotify: All classifiers exceeded the 20% random baseline but remained below 40% due to feature poverty. The query label reflects the Spotify search term used to retrieve each track, not an intrinsic audio property. The remix class was the

most distinguishable (F1=0.61). The prior work gap (−10%) directly quantifies the contribution of audio features.

- Housing: Linear and Ridge Regression both achieved $R^2 \approx 0.65$, closely matching the prior work benchmark of 0.678. The Decision Tree consistently underperformed ($R^2=0.44$), corroborating the prior study's conclusion that linear models are optimal for small housing datasets with binary features. Residual heteroscedasticity at high price points indicates that log-transforming the target would improve all models.

Cross-Dataset Model Comparison

Model Family	Heart Disease	Spotify (Clf)	Housing	Consistency
Logistic Regression	80.3% Acc	35.2% Acc	$R^2=0.650$ (Linear)	Strong on linear problems
Decision Tree	70.5% Acc (worst)	36.9% Acc	$R^2=0.435$ (worst)	Prone to overfitting on small data
k-NN	82.0% Acc (best)	37.3% Acc (best)	$R^2=0.584$	Best classifier; weaker on regression
Naive Bayes / Ridge	78.7% Acc	N/A	$R^2=0.649$ (Ridge=best)	Robust probabilistic/regularized baselines

- Strengths: Three diverse datasets demonstrating classification and regression pipelines; consistent GridSearchCV tuning; appropriate metrics per task type (AUC for medical, R^2 for regression); cross-dataset literature comparisons providing meaningful benchmarks.
- Limitations: Small sample sizes (545–1,025 rows for HD/Housing) increase variance. No deep learning or ensemble methods (Random Forest, Gradient Boosting) were applied. Spotify lacks audio features. Housing lacks geographic data. Heart Disease combines four UCI sub-databases, which may introduce batch effects.

Conclusion

Dataset	Task	Models	Best Model	Best Score	Prior Work Benchmark
Spotify	5-class Clf	LR, DT, k-NN	k-NN (k=31, manhattan)	37.3% Acc	47.3% (audio features)
Spotify	Duration Reg	LinReg, DT Reg	DT Regressor (tuned)	$R^2=0.077$	N/A

Heart Disease	Binary Clf	LR, DT, k-NN, NB	k-NN (k=21, manhattan)	81.97% / AUC 0.895	99% (deep learning RF)
Housing	Regression	LinReg, Ridge, DT, k-NN	Linear Regression	$R^2=0.650$	$R^2\approx0.678$

Across all three datasets, shallow classifiers performed competitively with or slightly below prior work benchmarks. The primary sources of performance gaps were the absence of audio features (Spotify), the use of deep learning ensembles in prior work (Heart Disease), and dataset composition differences (Housing). Decision Trees consistently required careful pruning to prevent overfitting on small datasets, while k-NN and linear models generalized more reliably. Future improvements across all datasets would benefit from ensemble methods (Random Forest, Gradient Boosting), expanded feature sets, and log-transformation of right-skewed regression targets.

References:

Mansisu08 (2023). Spotify Tracks Dataset. Medium.

https://medium.com/@mansisu08/spotify-tracks-dataset-abf6d02cdc49

PMC (2025). Deep Learning Models for Predicting Heart Disease Risk Using the UCI Database. PubMed Central

https://pmc.ncbi.nlm.nih.gov/articles/PMC12816779/

Madhuri, C.R. et al. (2019). House Price Prediction Using Regression Techniques. IEEE ICSSS

https://www.researchgate.net/publication/336876563_House_Price_Prediction_Using_Regression_Techniques_A_Comparative_Study